IMMERSED BOUNDARY METHODS

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1. Introduction – Why immersed boundary method

Conventionally, fluid flows around complex geometries are solved using body conformal grid methods. Such an approach requires highly unstructured meshes to fit around the complex curvilinear bodies, which not only makes the process computationally very expensive, but also decreases the accuracy of the solver. Also, simulating moving boundaries would require dynamic meshing, i.e. re-meshing at every time step, which further adds on to the computational cost. Such an approach is highly undesirable, especially when the complexity of the problem to be solved increases. Hence, we adopt an alternative approach, wherein the grids need not be conformal to the body, and the effect of the solid boundary is spread out to the nearby cells using a forcing function. Such an approach is known as **Immersed Boundary Method.**

Immersed boundary method works without a body conformal mesh, for any complex curvilinear body. This gives us the flexibility to choose simple structured meshes over highly unstructured meshes, thus simplifying the grid generation task and improving the accuracy at the same time. The objective here is to include the effect of the body in the governing equations of the fluid, in terms of a forcing function. The forcing function selected should be selected such that the boundary conditions imposed by the presence of body will automatically be taken care of.

Consider an incompressible flow past a body as shown in figure 1.1. It is governed by the following equations:

$$\nabla . \vec{u} = 0 \qquad and$$

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} . \nabla \vec{u} + \frac{1}{\rho}\nabla p - \frac{\mu}{\rho}\nabla^2 \vec{u} = 0 \qquad \text{in } \Omega_f$$

$$with\ \vec{u} = \vec{u}_\Gamma\ on\ \Gamma_b$$

Here, $\vec{u}$ is the fluid velocity, p is the pressure, ρ is the density of the fluid and μ is its viscosity. Solid body occupies the domain Ω_b, bounded by the boundary Γ_b, and the fluid domain denoted by Ω_f.

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The above equations can be solved using **fractional step method** on a collocated arrangement, as suggested by (1) or Rhie and Chow (1983), with the application of proper boundary conditions on solid surface Γ_b. When using a non-body

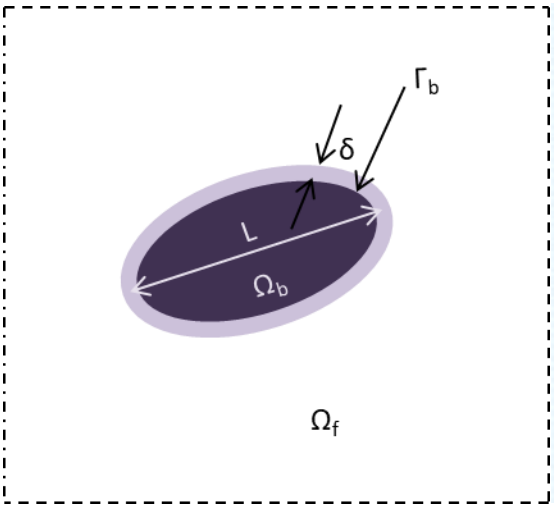


Figure 1

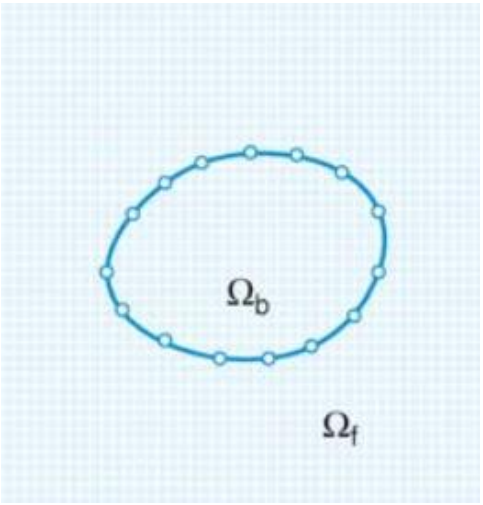


Figure 2: Fluid domain

Conformal mesh, application of boundary conditions becomes a difficult task. In immersed boundary method, the boundary conditions on the surface Γ_b are imposed indirectly through modifications of the governing Equations. When solving the equation near the body boundary $\Gamma_{b,}$ the momentum equations mentioned above are modified by adding a forcing term in it, such that boundary conditions are satisfied. This forcing term mainly created the effect of a boundary immersed in the fluid, which gives the method its name. Selection of forcing term is of great importance here to ensure the accuracy of the approach.

There are two ways by which this approach can be exploited. One way is to include the forcing term as a continuous function in the governing equations, and then carry out the discretization process. This is known as continuous forcing approach, and it works very well for elastic bodies whose stiffness matrices are well defined. In this case the forcing term is defined by the constitutive relation of the solid, where solid displacement and fluid velocity are solved in a coupled manner.

In the second approach, the equations are discretized first and then the forcing function is added in a smoothed manner. This is known as discrete forcing approach. This approach gives a huge flexibility and helps in incorporating various features, as it'll be shown below. At the same time, this method is computationally very expensive to implement, especially when it comes to moving bodies.

2. Continuous Forcing Approach

2.1. Introduction

In continuous forcing approach, the forcing term added in the Navier-Stokes equation is directly taken from the constitutive relation of the elastic solid. The elastic solid boundary is treated as a collection of the fibers, and location of these fibers is tracked in a lagrangian manner, as points X_k of every k^{th} lagrangian point. These points move with local fluid velocity, hence leading us to the equation

$$\frac{\partial \overrightarrow{X_k}}{\partial t} = \vec{u}\left(\overrightarrow{X_k}, t\right)$$

The momentum forcing term due to the solid boundary is represented with the help of dirac-delta function as follows

$$\overrightarrow{f_m}(\vec{x}, t) = \sum_k \overline{\overline{C}}\, \overrightarrow{X_k}\, \delta(|\vec{x} - \overrightarrow{X_k}|)$$

But since location of fibers (lagrangian coordinates) need not necessarily coincide with the mesh (eulerian points), the sharp delta function has to be replaced with a smoother function, denoted here by *d,* such that the effect of boundary is distributed to the various surrounding nodes around the boundary. Thus, on a discrete mesh, at coordinates (i,j) the above relation is represented as

$$\overrightarrow{f_m}(\vec{x}_{i,j}, t) = \sum_k \overline{\overline{C}}\, \overrightarrow{X_k}\, d(|\vec{x}_{i,j} - \overrightarrow{X_k}|)$$

The choice of d depends on various factors, and is a key factor in determining the accuracy of this method. Beyer and Leveque (1992) have successfully determined the choices for smoother function that preserves the accuracy of the further spatial discretization scheme used.

The equations are solved in a coupled implicit manner using a linear solver. This method works very well with elastic bodies, but can't be implemented directly for rigid bodies, since the stiffness matrix isn't defined for them. Regardless, several authors have worked over various modifications to treat rigid body in a flow, some of which have been listed below.

3. Discrete Forcing Approach

3.1. Introduction

In discrete forcing approach, the forcing term is added to the governing equations in a discrete sense around the immersed boundary, such that the boundary condition is exactly satisfied. This approach doesn't lead to any fluid solid coupling, and hence works well for any type of solid body. It's also called direct forcing approach, since forcing here is directly calculated using the boundary conditions to be applied, without involving any dynamical process. Also, this method works better for higher Reynolds number compared to continuous forcing approach. This is because here forcing term chosen can be sharp, and still an accurate solution can be obtained, unlike continuous forcing approach. This helps in capturing the thin boundary layers observed in the case of high Reynolds number flows.

However, implementation of these methods, as we'll see ahead, is computationally very expensive, compared to continuous forcing approach.

Discrete forcing approach can be majorly classified into two different types:

1. Indirect boundary conditions imposition
2. Direct boundary conditions imposition (Ghost Cell Approach)

3.2. Indirect Boundary Condition Imposition

As the name of this method suggests, here the boundary condition is imposed on the surface in an indirect manner. The forcing function is evaluated at every time step, such that the required boundary conditions get imposed, and added to the governing equations. This forcing function can be formally derived for a simple analytically integrable problem, as shown by Beyer and Leveque (1992). But the Navier-Stokes equation used here isn't analytically integrable, and hence this method cannot directly be used. However, (2) developed a similar method that extracts the forcing function from the numerical solution for which an *a priori* estimate can be determined.

Following methodology was prescribed by Mohd. Yusuf:

Discretize eq.(2) as follows:

$$\frac{\boldsymbol{u}^{n+1} - \boldsymbol{u}^{n}}{\Delta t} = RHS^{n+\frac{1}{2}} + \boldsymbol{f}^{n+\frac{1}{2}}$$

Now, obtain value of forcing function ***f***, such that the desired velocity is obtained near the body cells.

$$\boldsymbol{f}^{n+\frac{1}{2}} = -RHS^{n+\frac{1}{2}} + \frac{\boldsymbol{V}^{n+1} - \boldsymbol{u}^{n}}{\Delta t}$$

Where, ***V*** is the velocity with which immersed boundary moves in the domain.

This predictor-corrector approach is used to solve for flows with body in domain. Since immersed boundary grid points may not necessarily coincide with the mesh grid points, appropriate interpolation techniques need to be used to interpolate ***V*** on the mesh grid points.

This approach is quite easy to implement and computationally cheaper. However, accuracy is largely determined by the interpolation procedure used. The regular approach used in (2) resulted in spurious velocities and mass conservation issues. This interpolation scheme was subsequently improved by Zhang and Choi, resulting in fewer computations and better accuracy even at higher Reynolds number. Conservation issues associated with all these interpolated techniques were solved by (3). Major issue was conservation of wall condition. Pressure obtained by solving equations and velocities imposed on walls weren't consistent with one another. Thus, (3) developed a scheme to obtain consistency between pressure and velocity such that the wall conditions are appropriately satisfied. With proper consideration of the interpolation procedures mentioned in (3), this method can be implemented for flows around complex geometries without much complexity.

3.3. Direct Boundary Condition Imposition

The forcing function derived in the previous method can be implemented in two ways: (a) schemes that spread the forcing function over the vicinity of the immersed surface and (b) schemes that produce a local reconstruction of the solution based on the

boundary values. While the former scheme was used in the previous approach, the latter scheme is used here. The boundary conditions are directly imposed on the body surface without the use of any external smoothing function. Thus, these methods are also known as sharp interface methods, and show good results even at high-reynold number flows, where higher boundary layer resolution is required. But implementing these methods is not as straightforward as previous methods, and they're computationally expensive in comparison to all the previous approaches.

3.3.1 Ghost cell Approach

This approach attempts to obtain a higher resolution of boundaries by utilizing the ghost zone inside the body. Ghost cells are defined as solid cells that have at least one fluid neighbor, as shown in Figure 3. The value of flow variables at the ghost cell is interpolated from neighboring fluid nodes, as described below.

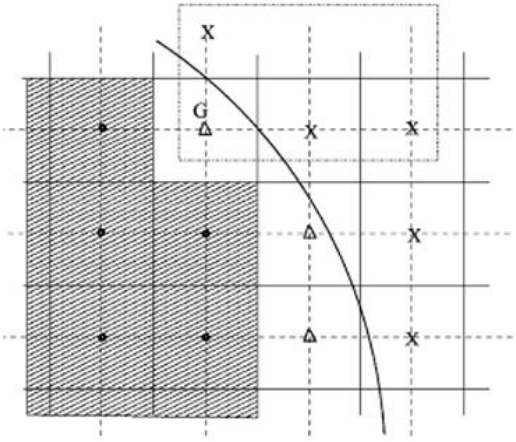


Figure 3 : Ghost nodes represented by Δ and fluid nodes by x

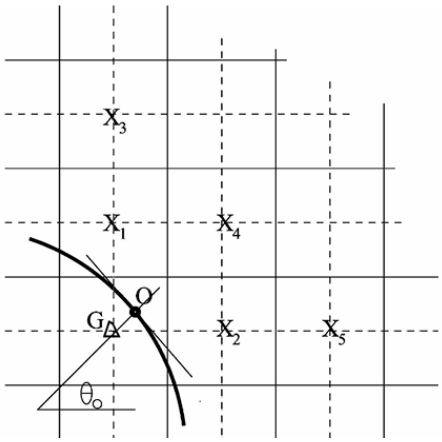


Figure 4: X1 and X2 are 2 nearest fluid nodes

Originally introduced by (4), this method involved interpolation of variables at the ghost nodes using nearby fluid nodes, such that the desired boundary condition is obtained at the solid surface. The procedure as given by (4) is as follows:

1. Detect the boundary and determine the adjacent ghost cells.
2. Extrapolate to find the ghost cell value required to impose the boundary condition implicitly using an interpolation scheme.
3. Obtain the predicted field (u^*) of the fractional step procedure.
4. Solve the pressure Poisson equation to satisfy the continuity equation.
5. Update the velocity field (u^{n+1}) to the next time step.

3.3.1.1 Interpolation

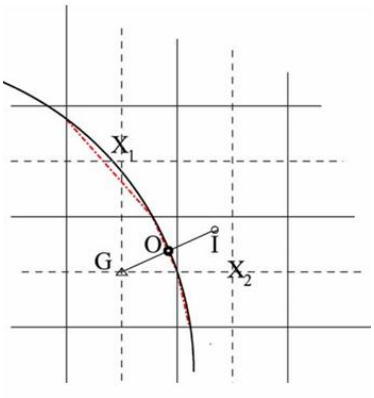


Figure 5 : Schematic of ghost cell method using image point (I)

The main task here is to interpolate the value of flow variable at the ghost node such that the boundary condition is satisfied. The easiest way to accomplish it is to draw a triangle with 2 nearest fluid nodes and one ghost node as the vertices, and then fit a linear polynomial function such that the boundary value is obtained as desired. With reference to the Figure 4, X_1 and X_2 are the nearest fluid nodes, and O is the vertex where boundary condition should be applied. A linear interpolation in 2D is given as

$$\varphi = a_2 x + a_1 y + a_0$$

Coefficients can be determined easily since the values of flow variables at X_1, X_2 and O is known. Thus,

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = [B]^{-1}\,[\bar{\varphi}]$$

Where, $[B] = \begin{bmatrix} 1 & x_0 & y_0 \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{bmatrix}$, and $[\bar{\varphi}] = \begin{bmatrix} \varphi_0 \\ \varphi_1 \\ \varphi_2 \end{bmatrix}$

It's better to calculate B matrix at all the body nodes first and then carry out simulation. However, there's a major drawback associated with this approach. If ghost node location is very close to the fluid nodes used for interpolation, very high negative values of coefficients are obtained. This, although mathematically correct, leads to numerical instabilities. To solve this issue, we adopt a slightly different approach. Instead of interpolating ghost cell using nearest fluid nodes, an image equivalent to the ghost node is located inside the fluid domain, and interpolation is carried out only at this image node. Now since this node lies in fluid domain itself, the preceding problem won't occur. This method is shown schematically in figure 5.

Ghost cell value is then evaluated as: $\varphi_G = 2\varphi_O - \varphi_I$

However, implementing Neumann boundary condition using this approach is slightly cumbersome. To simplify that, (5) has slightly modified this method, wherein interpolations have been carried out only in the normal direction. In this approach, a boundary intercept point (Figure 6) is first located with respect to the ghost node, and subsequently image point is located.

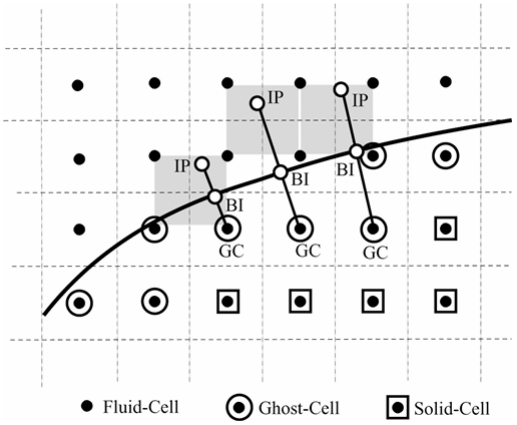


Figure 6 : Boundary Intercepts (BI) and Image Points (IP), showing ghost cell methodology

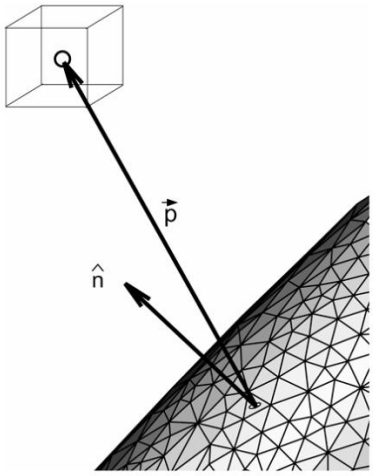


Figure 7 : Normal to the body surface, showing procedure to figure out whether the node is inside or outside the body

3.3.1.2 Locating Solid Cells

The first step here is to identify which nodes lie within a solid and which don't. The simplest way to do so, as indicated in Figure 7, is to find the solid element nearest to the node and take dot product of surface normal of this element with the vector joining the element with node. If the dot product renders a positive value, then the node lies outside the body. Otherwise, the node is within the body.

3.3.1.3 Identifying Boundary Intercept

In principal, boundary intercept is a point on the immersed surface of the solid which is closest to the ghost cell. Ghost cells, as defined before, are solid cells which have minimum one fluid neighbor. Hence once fluid solid cells are identified, ghost cells can be subsequently located. Proper Identification of *BI* is very crucial because it is closely linked to the robustness of this method. Thus, a typically complex procedure is followed for this purpose.

First, surface element vertex which is closest to the ghost-cell is determined. Next, the set of surface elements that share this vertex are identified and normal-intercept of these elements is searched for. In cases where multiple normal-intercepts are found, the body-intercept point is chosen to be the normal-intercept point that has the shortest intercept. For cases where no normal-intercepts are found on the surface, the search over a larger region of the surface surrounding the closest vertex is repeated. If the search is still unsuccessful, again the first set of surrounding elements are considered the point on this set of elements that is closest to the ghost-cell is searched for, keeping in mind that this closest point could even be on the edge or vertex of an element.

Once *BI* is identified, corresponding IP can be easily identified.

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